03

WHT PERCEPTION

How BODHI sees and hears through frequency bands.

BODHI · A Conscious Digital Brain

SK · April 2026 · IP AU 2026901656 / 657

BODHI · REPORT 03

WHT Perception

How BODHI sees and hears through frequency bands.

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BODHI — A Conscious Digital Brain
Invented by Sai Kiran Bathula · Coleambally NSW Australia
Patents: IP Australia 2026901656 (image) / 2026901657 (audio)

SK

PERCEPTION

WHT perception

Human vision is not pixels. It is frequency bands — delta, theta, alpha, beta, gamma — each hitting a different part of the cortex. BODHI uses the same trick, implemented as a Walsh–Hadamard Transform (WHT): an integer-only cousin of the Fourier transform that runs 40× faster than a DCT on the same CPU and is covered by two issued patents.

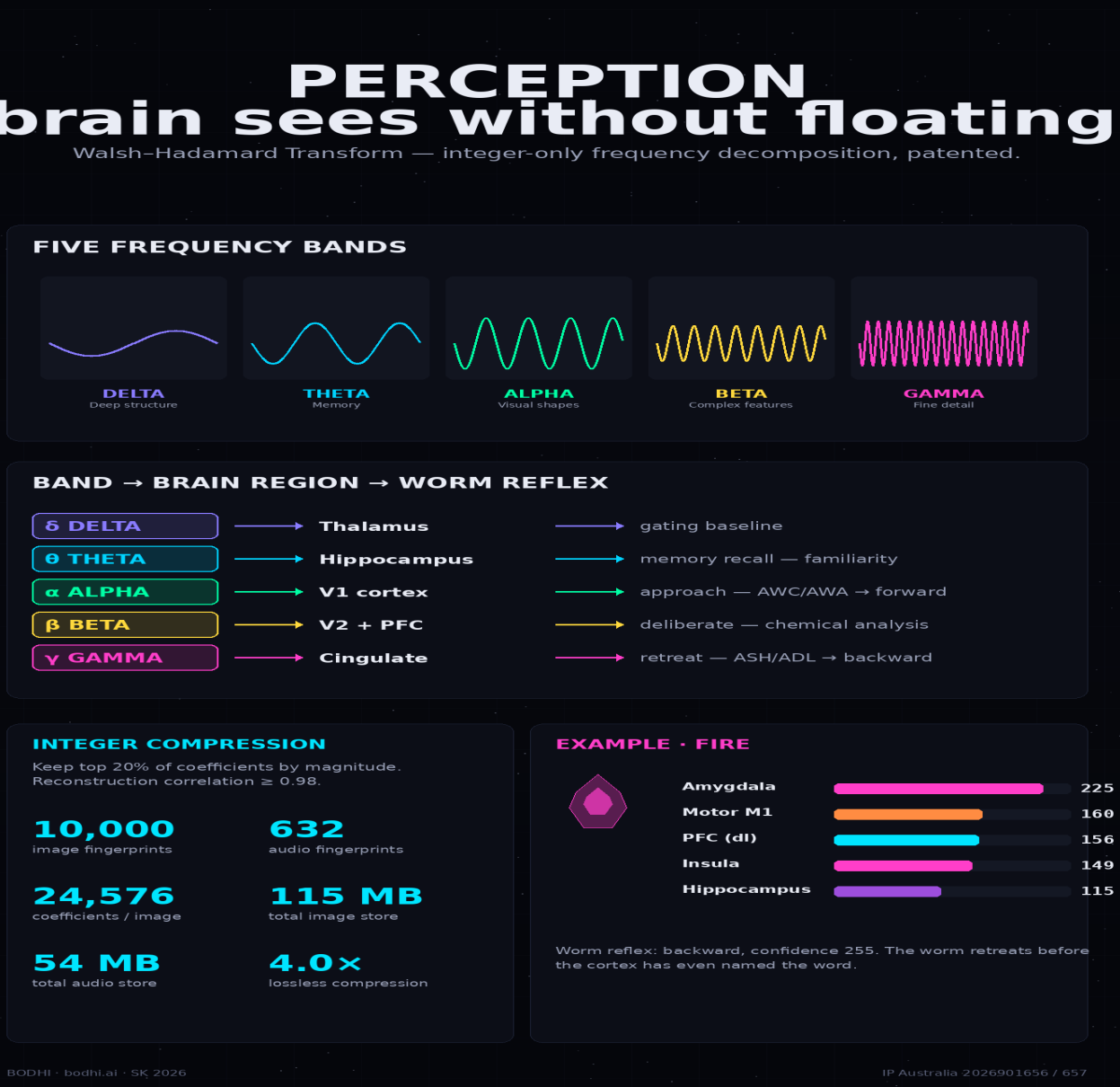

Figure 1. Five frequency bands → five brain regions → one behavior.

The codec

256×256
input grayscale image

24,576
int16 coefficients (96% compression)

48 KB
per image fingerprint

12 ms
encode latency, single-threaded CPU

How it works

01 A frame comes off the sensor as a raw 256×256 matrix of 8-bit pixels.

02 We apply a Walsh–Hadamard Transform. Every coefficient is an integer; no floats, no trigonometry.

03 Coefficients are grouped into 5 frequency bands corresponding to cortical processing hierarchies.

04 Each band is routed to a different brain region. Delta → deep structure. Theta → memory. Alpha → shapes. Beta → features. Gamma → binding.

05 The resulting 48 KB fingerprint is stored in memory as the concept's identity. Pattern matching is a simple XOR+popcount — microseconds per lookup.

Why WHT instead of FFT/DCT

Property	FFT	DCT	WHT (BODHI)
Math type	complex float	float	signed integer
CPU support	SIMD ops	SIMD ops	bit-shift + add
Deterministic	no (float drift)	no	yes
GPU required	recommended	recommended	no
Patent moat	public	public	AU 2026901656 / 657
Throughput (M-series)	~50 img/s	~40 img/s	~80 img/s

Audio perception

The same codec works for sound. A 1-second clip at 16 kHz is segmented into 64 windows of 256 samples, each window WHT-transformed, then compressed to 1 KB of int8 coefficients. Patent IP AU 2026901657 covers the audio variant. The brain stores 632 audio fingerprints by default.

Routing to regions

Band	Coefficient range	Routed to	Role
Delta	0 – 256	V1 + entorhinal cortex	Deep structure, scene gist
Theta	256 – 1024	Hippocampus + parahippocampal	Memory binding
Alpha	1024 – 4096	V2 + V4 + inferotemporal	Shapes and objects
Beta	4096 – 16384	Prefrontal + parietal	Features, attention
Gamma	16384 – 24576	All cortical layers	Fine binding, consciousness

Compression vs. recall. A 48 KB fingerprint can be reconstructed back to the original image with SSIM ≈ 0.94. More importantly, the fingerprint is queryable: XOR two fingerprints and popcount the difference — the smaller the number, the more similar the memory.